STACKS, DIFFERENTIAL GRADED SHEAVES: LEMMA-QIS

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 4369-4379 of sources/stacks-sdga/pinned-original-source.tex; proof span is lines 4381-4411, the complete author proof (29 body lines).

Editorial note. Navigation stubs: 5 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. Queue index 22 of queue_sdga.json (pb=1311, stmt 4369-4379, proof 4381-4411).

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. The cited inputs are prerequisites; the argument in the retained span is not replaced by any cited result.

Source reference. Original lemma 04VC: sources/stacks-sdga/pinned-original-source.tex, source lines 4411–4411. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 05Q8: sources/stacks-sdga/pinned-original-source.tex, source lines 4411–4411. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 05R2: sources/stacks-sdga/pinned-original-source.tex, source lines 4411–4411. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 05R4: sources/stacks-sdga/pinned-original-source.tex, source lines 4411–4411. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0FT2: sources/stacks-sdga/pinned-original-source.tex, source lines 4316–4328. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Date: stacks-project:sdga.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Lemma 0.1. *Let $(\mathcal{C}, \mathcal{O})$ be a ringed site. Let $(\mathcal{A}, d)$ be a sheaf of differential graded algebras on $(\mathcal{C}, \mathcal{O})$. Consider the subclass $Qis \subset Arrows(K(Mod(\mathcal{A}, d)))$ consisting of quasi-isomorphisms. This is a saturated multiplicative system compatible with the triangulated structure on $K(Mod(\mathcal{A}, d))$.*

Proof. Observe that if $f, g : \mathcal{M} \to \mathcal{N}$ are morphisms of $Mod(\mathcal{A}, \mathrm{d})$ which are homotopic, then f is a quasi-isomorphism if and only if g is a quasi-isomorphism. Namely, the maps $H^i(f) = H^0(f[i])$ and $H^i(g) = H^0(g[i])$ are the same by Lemma 0FT2. Thus it is unambiguous to say that a morphism of the homotopy category $K(Mod(\mathcal{A}, \mathrm{d}))$ is a quasi-isomorphism. For definitions of "multiplicative system", "saturated", and "compatible with the triangulated structure" see Derived Categories, Definition 05R2 and Categories, Definitions 04VC and 05Q8.

To actually prove the lemma consider the composition of exact functors of triangulated categories

$$K(Mod(\mathcal{A}, \mathrm{d})) \longrightarrow K(Mod(\mathcal{O})) \longrightarrow D(\mathcal{O})$$

and observe that a morphism $f : \mathcal{M} \to \mathcal{N}$ of $K(Mod(\mathcal{A}, \mathrm{d}))$ is in Qis if and only if it maps to an isomorphism in $D(\mathcal{O})$. Thus the lemma follows from Derived Categories, Lemma 05R4. □